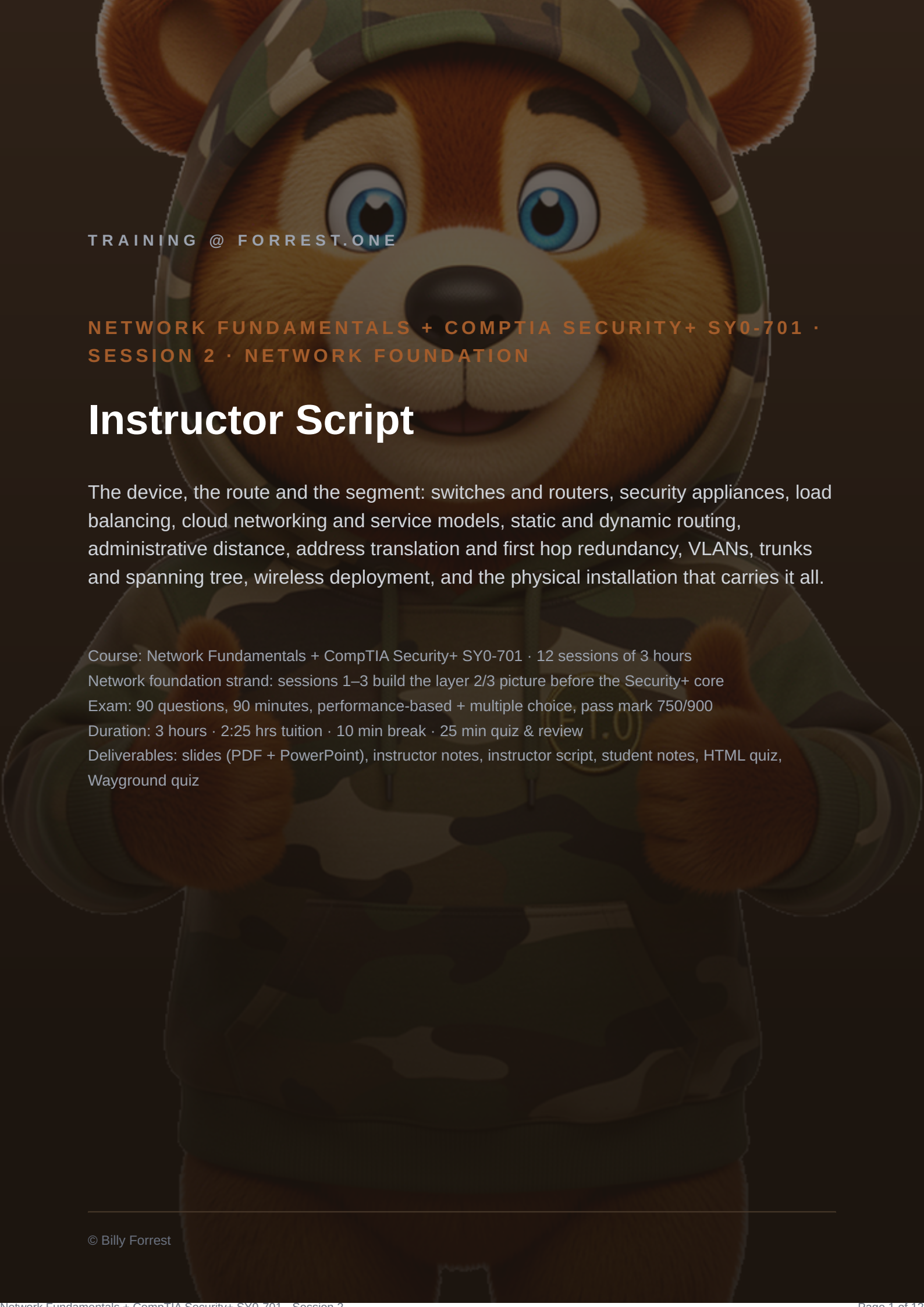




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 2 · NETWORK FOUNDATION**

Instructor Script

The device, the route and the segment: switches and routers, security appliances, load balancing, cloud networking and service models, static and dynamic routing, administrative distance, address translation and first hop redundancy, VLANs, trunks and spanning tree, wireless deployment, and the physical installation that carries it all.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Script — Switching, VLANs & Network Architectures

A word-for-word delivery script for Session 2, following the expanded session narration frame for frame. Stage directions appear in square brackets. Each slide runs roughly 40–90 seconds of speech; the two discussions and the segmentation activity expand the timings in the instructor notes. The script follows five chapters: infrastructure devices, cloud and routing foundations, route selection and translation, VLANs and spanning tree, then wireless and physical deployment.

Slide 1 · Title

Good day, and welcome to Session Two. Session One gave you the models and the numbers. Now we put real devices around them. We open with switches and routers, then the security appliances that guard the traffic: firewalls, intrusion detection and intrusion prevention. Then load balancers, proxies, storage and wireless hardware. From there we move to the cloud, to routing, to virtual local area networks, or VLANs, and to wireless deployment. One thread runs through it all: every device is also a security boundary.

Slide 2 · By the end of this session you will be able to

By the end of this session you will do ten things. Compare a switching decision against a routing decision. Describe firewalls, intrusion detection and intrusion prevention. Explain load balancing and proxy servers. Compare network attached storage against a storage area network. Define cloud virtualisation, private networks, security groups and gateways. Compare the cloud deployment and service models. Compare static, distance vector, link state and path vector routing. Explain administrative distance and the longest match rule. Describe address translation, first hop redundancy, VLANs and wireless security. And finally, build and defend the fabric itself. Ten outcomes, and the quiz at the end tests every one.

Slide 3 · Network Infrastructure Devices

Chapter one is about devices. Switches and routers first: layer two frame forwarding against layer three path selection. Then the security appliances, the firewalls, the intrusion detection and the intrusion prevention systems. Then the traffic shaping devices, the load balancers and the two kinds of proxy. Then storage and wireless hardware, network attached storage, storage area networks, access points and controllers. And then cloud networking, where virtualised network functions, private clouds and service models change what a device even is. [Pause.] Ask of each one, who does this protect, and from whom? Detection alone is not defence.

Slide 4 · Switches forward frames, routers forward packets

Start with the two workhorses. A switch works at layer two and forwards frames by media access control, or MAC, address. That address is forty eight bits written as six pairs of hexadecimal digits, and the first three octets are the organisationally unique identifier, the OUI, which names the manufacturer. The switch builds its table by learning the source address on every arriving frame. A router works at layer three and forwards packets by Internet Protocol, or IP, address. The difference in scope is the whole point. A switching decision is local to one network, and a routing decision joins separate networks. Routers learn those paths with routing protocols such as Open Shortest Path First and Border Gateway Protocol, and the routing table records a next hop for each destination.

Slide 5 · The switch forwarding table

Four behaviours, in order. Learn: record the source hardware address and the port it arrived on. Look up: find the destination address in the table. Forward: send the frame out of the one port that holds that destination. Flood: if the destination is unknown, send it to every port except the one it came in on. Broadcasts are always flooded, and that is exactly why a loop is fatal. Now the security consequence. Fill that table with bogus source addresses and the switch runs out of room, degrades to hub mode, and sends every frame everywhere. That is MAC flooding. It is dangerous precisely because it undoes segmentation at the port, and port security is the control that stops it.

Slide 6 · Security appliances: detect, prevent, filter

Now the guards. A firewall filters traffic against rules built from addresses, port numbers, protocols and content. It sits at the perimeter, and internally to separate zones. An intrusion detection system, or IDS, monitors passively and raises an alert, but never blocks anything. An intrusion prevention system, or IPS, sits inline and can drop the packet, block the source or reset the connection. A network sensor watches a whole segment; a host sensor watches one machine. Together that is defence in depth. [Pause.] Say it plainly: the firewall refuses by rule, detection reports after the fact, and prevention stops it in line. An alert with nobody watching it is not a control.

Slide 7 · Load balancers and proxies

Two devices that shape traffic. A load balancer spreads requests across a pool of servers, so no single server is overwhelmed. Round robin sends each request to the next server in turn, and least connections picks the server with the fewest active sessions. If one server fails, the pool keeps serving. A forward proxy hides the client and filters outbound requests. A reverse proxy sits in front of the servers, caches content and terminates transport layer security, or TLS. Both hide one side of the conversation: the forward proxy protects the client side, the reverse proxy protects the server side.

Slide 8 · Storage and wireless hardware

Storage and wireless hardware. Network attached storage, or NAS, serves files over the network at file level, using protocols such as Network File System and Server Message Block. A storage area network, or SAN, serves blocks at high speed over Fibre Channel and appears to the server as a local disk. An access point bridges wireless clients onto the wired network. A wireless controller configures and monitors many access points from one policy interface, and a wireless bridge joins two segments with no cable between them. [Pause.] Learn each pairing, because the exam asks which one a server actually needs.

Slide 9 · Discussion

[Run the discussion per the instructor notes: six minutes in pairs, four minutes of debrief.] A firewall, a detection system and a prevention system all see the same traffic. Which one stops an attack? The prevention system, because it is inline and can drop the packet. Why can you not rely on it alone? Because signatures lag, tuning is imperfect, encrypted traffic limits inspection, and the detection system and the firewall rule set cover the gaps. Ask the room who reads the alerts. If nobody owns the alert queue, defence in depth is a diagram rather than a control.

Slide 10 · Cloud Networking and Routing Foundations

Chapter two turns to the cloud, and then to routing. Cloud computing replaces dedicated appliances with software, and we will see what that changes for security. Then the routing story begins: a static route typed by hand, and a dynamic protocol that learns on its own. The thread from chapter one continues. Watch where the attack surface sits, because it moves with every design choice. [Pause.] Ask of each option, who does this protect, and from whom? Every one of these controls decides who reaches what.

Slide 11 · Network functions virtualisation

Cloud networking changes what a device is. Network functions virtualisation, or NFV, replaces dedicated hardware appliances with software on ordinary servers. A firewall, a router or a load balancer becomes a virtual machine. That lowers cost, speeds deployment and lets you scale up or down on demand: a virtual firewall deploys in a moment, not after a procurement cycle. The trade is that the appliance you once racked and patched by hand is now software. Its management plane and its hypervisor become the things an attacker goes after. The function is the same. The boundary moved.

Slide 12 · Private clouds, security groups and gateways

Inside a public cloud you build a virtual private cloud, or VPC. That is a logically isolated network where you choose the address ranges, the subnets and the route tables. Traffic is controlled by security groups, which are stateful firewalls attached to a resource, so return traffic is allowed automatically. Network access control lists filter at the subnet boundary instead. A virtual private network gateway gives an encrypted tunnel back to the office, and a direct connection is private and avoids the public internet entirely. [Pause.] Default to deny, then open only what the service actually needs.

Slide 13 · Deployment and service models

Deployment and service models. A public cloud shares provider infrastructure and bills as you go. A private cloud is dedicated to one organisation, and suits strict compliance needs. A hybrid keeps sensitive data private and puts the rest in the public cloud. Service models shift the boundary of responsibility. With software as a service the provider runs the whole application, and you manage only your data and your users. With infrastructure as a service you also run the operating system. With platform as a service you build only the application. [Pause.] The responsibility shifts. It never disappears.

Slide 14 · Static routing

Routing begins with a decision you make by hand. A static route is entered by an administrator and stays exactly as written. That suits a small network, a stub network with one path in and out, or a default route that catches every destination the table does not list. Static routes carry an administrative distance of one, so a router trusts them above anything learned dynamically. The cost is that they do not adapt. If the link fails, the route stays, and traffic is sent into a dead end.

Slide 15 · Dynamic routing and distance vector

Dynamic protocols remove the hand typing. A distance vector router sends its whole routing table to its neighbours, and they pass it on, which is routing by rumour. Routing Information Protocol, or RIP, measures a path in hops. Fifteen hops is the maximum usable count, and sixteen means unreachable, which is how RIP stops a route circling for ever. Convergence is slow, and while routers disagree you get loops and black holes. Split horizon, route poisoning and hold down timers are the classic defences. Plan for the failure.

Slide 16 · Open Shortest Path First

Open Shortest Path First, or OSPF, is the open standard link state protocol. Each router describes its own links in a link state advertisement, and those advertisements flood to every other router in the area. Every router then holds the same link state database, which means the same map of the network. Each router runs the shortest path first calculation, also called Dijkstra, with itself as the root. Cost follows bandwidth, so a lower cost is preferred. Area zero is the backbone, and every other area attaches to it. [Pause.] Distance vector routers trust rumour. Link state routers hold the map.

Slide 17 · Route Selection, Translation and Inter-VLAN Routing

Chapter three is about choosing and translating. When two sources offer a route to the same destination, administrative distance decides which source the router trusts. Then we rewrite the addresses themselves as traffic crosses the boundary. Then we make the default gateway survive a router failure. And finally, one physical link carries every VLAN. Each of these is a control you may be asked to justify. Ask what breaks if the mechanism is misconfigured, because something always does. Misconfiguration here is a very common real outage.

Slide 18 · EIGRP and BGP

Two more routing protocols. Enhanced Interior Gateway Routing Protocol, or EIGRP, is advanced distance vector. Its diffusing update algorithm keeps a feasible successor, which is a backup path already calculated. When the primary route fails, that backup takes over without recalculating the whole network, so recovery is quick. It weighs bandwidth, delay, load and reliability rather than hop count alone. Border Gateway Protocol, or BGP, is the path vector protocol of the internet, and it selects on attributes and policy, such as the length of the autonomous system path. BGP is also the protocol behind route hijacks, so both must be secured and monitored.

Slide 19 · Administrative distance decides, then the metric

A router often learns the same destination from more than one source. Administrative distance settles the argument, because it ranks how trustworthy each source is, and the lowest value wins. A connected route scores zero, so it always wins. A static route scores one. EIGRP scores ninety, OSPF scores one hundred and ten, and RIP scores one hundred and twenty. Once a source is chosen, metrics pick between its paths. And the longest match rule decides first of all, so the most specific route wins. [Pause.] Trust order first, then the metric.

Slide 20 · Network address translation

Network address translation, or NAT, rewrites addresses in the packet header as traffic crosses a router or a firewall. It exists for two reasons: to stretch the shrinking pool of IPv4 addresses, and to hide the internal layout from the outside world. Static NAT maps one private address to one public address, which is what a published web server needs. Dynamic NAT takes an address from a pool as required. The vocabulary is inside and outside for position, and local and global for scope. [Pause.] Log the translations. A translation table is evidence and a troubleshooting tool at the same time.

Slide 21 · Port address translation

Port address translation, or PAT, is the version you meet every day. It maps many private addresses onto one public address, and tells the sessions apart by port number. When a device opens a connection, the router rewrites the private address and port to the public address and a unique port. That mapping is held in the translation table. Two hosts can then appear as one public address on two different ports. Because the router cannot tell who should receive an unsolicited connection, inbound services need port forwarding.

Slide 22 · First hop redundancy protocols

If the default gateway dies, everything behind it loses the outside world. First hop redundancy protocols fix that by giving several routers one shared virtual IP address. Hot Standby Router Protocol, or HSRP, keeps one active router and one standby. Virtual Router Redundancy Protocol, or VRRP, is the open standard version, with a master and several backups. Gateway Load Balancing Protocol, or GLBP, goes further and

shares the forwarding load. Preemption hands the role back to the best router, and the client keeps the same gateway address throughout. [Pause.] Test the failover before you need it.

Slide 23 · Subinterfaces and router on a stick

How does a router reach the VLANs sitting behind a switch? With subinterfaces. A subinterface is a logical division of one physical router port, and each one serves a single VLAN. The router connects to the switch over a trunk, and every subinterface carries the eight oh two point one Q tag for its own VLAN. That is the router on a stick design. It gives each VLAN its own gateway address, but all that traffic shares one physical link. A busy network will feel that bottleneck.

Slide 24 · Break

Ten minutes. When we return: the switching chapter. VLANs, trunks, port security, spanning tree, and the frame size that breaks networks quietly when it is wrong.

Slide 25 · VLANs, Switching and Spanning Tree

Chapter four is the switching chapter. A VLAN splits one switch into logical broadcast domains, and that is the core containment control of this session. Then trunks and tags, then port security, then spanning tree and its modern variants. Finally the frame size itself, which breaks networks quietly when it is wrong. [Pause.] Every one of these mechanisms exists for availability or for containment. And every one of them fails in a way that looks like something else at first. That is what makes this chapter worth your attention.

Slide 26 · VLAN fundamentals

A VLAN is a logical broadcast domain built inside the switching hardware. Without one, every device on the switch hears every broadcast. Devices join a VLAN by the port they use, by their hardware address, or by protocol. VLAN one is the default, and the native VLAN is the one that carries untagged frames. A separate management VLAN keeps administration traffic away from users. The payoff is containment: fewer broadcasts, a smaller blast radius, and no easy path for lateral movement. [Pause.] Name your VLANs so the design documents itself.

Slide 27 · Trunking and tagging

Between switches you need trunking. Eight oh two point one Q inserts a four byte tag into the Ethernet frame, and the VLAN identifier inside it is twelve bits. That gives just over four thousand usable VLANs. The receiving switch reads the tag and keeps the traffic in the right VLAN. One VLAN on the trunk is the native VLAN, and its frames cross untagged, which is why a native VLAN mismatch can leak traffic between VLANs. Access ports carry a single VLAN to one endpoint. Now the attacks: switch spoofing, where the attacker negotiates a trunk, and double tagging, where the native VLAN is spoofed. Tag every trunk consistently.

Slide 28 · A practical VLAN design

Here is a real design. VLAN ten, management, for switch and infrastructure administration. VLAN twenty, servers. VLAN thirty, workstations. VLAN forty, the DMZ, for public facing services. VLAN nine nine nine, the blackhole, an unused quarantine VLAN that swallows traffic from ports nothing should use. And the key rule. VLAN one is the default native VLAN on trunks, and it is a known attack surface. Move it to an unused ID, prune the VLANs you do not need, and restrict trunks to an explicit allowed list. That is what blunts VLAN hopping, where an attacker either spoofs a trunk or double tags a frame on the native VLAN to jump segments.

Slide 29 · Switch interface configuration

Now configure the port itself. Speed and duplex must agree at both ends. A duplex mismatch, where one side is full and the other half, produces collisions and terrible throughput. Port security caps how many hardware addresses may appear on a port, and a violation can drop traffic, log it, or shut the port down. PortFast skips the listening states on a port that leads to a single endpoint. BPDU Guard shuts a port that receives a bridge protocol data unit, which catches a rogue switch. [Pause.] A quiet access port should never speak spanning tree.

Slide 30 · Loops and the root bridge

Redundancy creates a problem. Two links between switches form a loop, and a broadcast circling that loop becomes a storm that consumes the network in seconds. The switch also sees the same hardware address on two different ports, so its table becomes unstable. Spanning tree protocol, or STP, prevents all of that by building one loop free logical topology. The switch with the lowest bridge identifier, which is priority plus hardware address, becomes the root bridge. Every other switch picks its best port toward the root, and the rest block. Classic convergence takes thirty to fifty seconds.

Slide 31 · STP port states and convergence

Five states. Blocking: no forwarding, listening for BPDUs, holding the link in reserve. Listening: transitional, still learning the topology. Learning: building the MAC table but forwarding nothing. Forwarding: normal operation. Disabled: administratively down. [Pause.] Blocking, listening and learning all mean no data. And the convergence note: traditional STP takes thirty to fifty seconds to converge after a topology change. Rapid spanning tree does it in seconds. Know which one your campus runs.

Slide 32 · Rapid and multiple spanning tree

Classic spanning tree is slow, so the standard moved on. Rapid spanning tree, defined in eight oh two point one w, converges in seconds rather than tens of seconds. It adds the alternate and backup port roles, and merges blocking with listening into one discarding state. Multiple spanning tree, defined in eight oh two point one s, maps groups of VLANs to instances so redundant links can actually be used. Per VLAN spanning tree

instead gives every VLAN its own tree. Whichever you run, choose the root bridge deliberately. Never leave the election to chance.

Slide 33 · STP attacks and the guard features

STP can be attacked. A rogue switch sending superior BPDUs can become the root bridge, and then all traffic traverses the attacker's switch, which is invisible interception rather than an outage. Three defences. Root Guard: a port protected by it cannot become a root port, so it neutralises a rogue root. BPDU Guard: an access port that receives a BPDU shuts down, stopping a rogue switch being plugged in at all. Port security: limits the MAC addresses allowed per port, which also blunts MAC flooding. [Pause.] The stakes are real. A rogue root silently reroutes your traffic.

Slide 34 · Discussion

[Run the discussion per the instructor notes.] Two switches, equal priority: which becomes root, and what converges? Equal priority, so the tiebreaker is the MAC address, and the lower MAC roots. On the non-root switch, the root port is the lowest-cost path, so the gigabit link at cost four forwards and the hundred megabit link at cost nineteen blocks. If the gigabit link fails, the blocked port transitions and the network reconverges: thirty to fifty seconds on traditional STP, seconds on RSTP.

Slide 35 · Maximum transmission unit and jumbo frames

One more setting that breaks networks quietly. The standard maximum transmission unit, or MTU, for Ethernet is one thousand five hundred bytes. A larger packet must be fragmented, and if the do not fragment flag is set it is dropped instead. When that happens, an Internet Control Message Protocol, or ICMP, message goes back to the sender. Jumbo frames raise the limit to about nine thousand bytes, which cuts overhead in data centres and storage networks. Every device in the path must agree on the size. [Pause.] An MTU mismatch is a classic cause of strange, partial failures.

Slide 36 · Activity: design the segmentation

[Run the activity per the instructor notes: six minutes in pairs, four minutes of debrief.] Which VLAN hosts the public web server? The DMZ. Which never carries user traffic? The blackhole. Why isolate management? Management should never share a broadcast domain with users, because a compromise there controls the switches. What does the default native VLAN risk? VLAN hopping through double tagging. And two attacks segmentation prevents: VLAN hopping, and the spread of a broadcast storm. [Pause.] Extend it: think microsegmentation, isolating sensitive workloads rather than whole departments.

Slide 37 · From segments to security zones

Now the architecture view. Segmentation, whether VLANs or subnets, becomes security zoning. The DMZ hosts public services, isolated from the internal network and bracketed by an external firewall and an internal firewall, often with a proxy between them. Blast radius: a compromise stays inside its segment. Lateral

movement: zones block east-west spread, so an attacker who lands on one workstation cannot roam the campus. Then the campus layers, access, distribution and core, and IoT isolation on its own segment. [Pause.] Architecture is a control, not plumbing.

Slide 38 · Wireless and Physical Deployment

Chapter five leaves the cable behind, and then comes back to it. We cover the frequency bands and channel widths, then the names and modes a wireless network uses. Then the encryption history, which is a lesson in what happens when a control is broken early. Then enterprise authentication and guest design. And finally the rack, the cable and the room. [Pause.] Notice how often the same principle returns: separate what must not touch. And notice how often physical access decides the final outcome.

Slide 39 · Channels and frequencies

Wireless starts with the air itself. The two point four gigahertz band has eleven channels, and only channels one, six and eleven do not overlap. Five gigahertz offers up to twenty three non overlapping channels, which is why busy offices prefer it. Six gigahertz, used by Wi Fi six E, adds far more spectrum again. Lower frequencies pass through walls better, and higher frequencies carry more data over shorter distances. Channel width matters too: twenty megahertz is the steady default, and wider channels raise throughput while consuming the space around them.

Slide 40 · Network names and modes

Now the names and the modes. The service set identifier, or SSID, is the network name that clients see. Infrastructure mode sends all traffic through an access point, and ad hoc mode lets devices talk directly, with weaker security and no central control. The basic service set identifier names a single access point, and is usually its hardware address. The extended service set identifier covers several access points sharing one name, which is what lets a client roam. [Pause.] Hiding the name stops nothing, and client isolation keeps strangers apart.

Slide 41 · Wireless encryption

Wireless encryption has a history you must know. Wired equivalent privacy, or WEP, was broken almost as soon as it appeared: a short key and a weak key schedule, and it falls quickly. Wi Fi protected access, or WPA, added the temporal key integrity protocol as a stopgap. WPA2 replaced that with the advanced encryption standard, or AES, and it is still the common baseline. WPA3 adds simultaneous authentication of equals, which resists offline cracking of a captured handshake. Personal mode shares one pre shared key, and enterprise mode uses eight oh two point one X with an authentication server. [Pause.] Retire WEP entirely.

Slide 42 · Enterprise authentication and guest networks

Enterprise wireless takes the decision away from the access point. Eight oh two point one X authenticates a device before it is allowed onto the network, and a central authentication server checks the credentials. Several methods carry that exchange under the extensible authentication protocol, or EAP. Protected EAP hides the exchange inside a TLS tunnel. EAP transport layer security goes further and uses certificates on both the client and the server. A guest network needs its own name, its own VLAN, a bandwidth limit and clear rules, and guests accept an acceptable use policy before they get even that. Log every authentication event.

Slide 43 · Access points and antennas

How access points are managed, and how their aerials shape coverage. A standalone access point is configured on its own, which is fine for a small site. A controller based access point is managed by a wireless controller, which sets power and channel for every access point and keeps policy consistent. An omnidirectional antenna radiates all around, for open areas. A directional antenna, such as a Yagi, focuses a beam for a long point to point link. Higher gain means more reach and a narrower pattern. [Pause.] A site survey and heat map show where coverage really lands. Survey before you mount.

Slide 44 · Installation, power and environment

Finally, the room itself. Equipment goes into a standard nineteen inch rack, and one rack unit of height is one point seven five inches, so a tall rack holds a great deal. Label every cable and document the patch panel, or troubleshooting becomes guesswork. A patch panel also saves wear on the switch ports. Lock the rack and control who enters the room, because physical access defeats every logical control you own. Cool the room with hot and cold aisles, back the power with an uninterruptible supply, and keep equipment away from water, heat and dust. [Pause.] Label it, or you will lose it.

Slide 45 · Quiz

[Direct candidates to Quiz_Session_02.html. Fifteen minutes.] Open the session quiz. Twenty questions, immediate feedback, and every question carries a topic tag so you can see what it tests. Read every explanation, because the distractors teach as much as the answers. Aim for seventy percent or better before you leave today. The Wayground version is available for in-class assessment, and every question in it maps to this deck.

Slide 46 · Key takeaways

Seven takeaways. One: switches forward frames by hardware address, and routers forward packets by IP address. Two: each appliance does one job, the firewall refuses, detection reports, and prevention blocks in line. Three: the cloud moves the boundary, with virtualised functions, isolated networks and stateful security groups. Four: administrative distance ranks the source of a route, the metric picks the path, and the longest match wins before either. Five: translation conserves public addresses, and first hop redundancy keeps the gateway alive. Six: VLANs contain, trunks carry the tags, and spanning tree keeps one logical path. Seven:

wireless means WPA2 or WPA3 with AES, a guest VLAN on its own segment, and an installation you can actually service.

Slide 47 · Revision checklist

Before next session: retake the quiz until you score seventy percent without guessing. Redraw one design from this session, a VLAN plan, an administrative distance table, or a wireless policy. Trace an STP election: two switches, two links, pick the root bridge and the blocked port. And read every device twice, once as plumbing and once as a boundary. Then read ahead to Session Three, where we meet the services every network runs: DNS, DHCP, NTP and SSH.

Slide 48 · Next session preview

Next time, Network Services and the Attack Surface. DNS, the directory that everything depends on, and the poisoning attacks against it. DHCP, automatic addressing, rogue servers and starvation. NTP, the clock that forensics rely on. And SSH, the encrypted management channel and how to read its logs. Every service is a service, and every service can be turned against you. [Pause.] Read every device twice. Thank you for your attention, and I will see you in Session Three.